ADVANCED OBJECT ORIENTED PROGRAMMING

(25CS2205E)

TRIMESTER 5

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JAVA LABORATORY OBSERVATION NOTES

Five Java class and object programming experiments with algorithms, source code, output screenshot spaces and results.

Experiment 1: Student Class

Aim

To create a Student class with student ID, name and age, create three student objects and display their details.

Software Requirements

- MacBook / macOS
- Visual Studio Code (VS Code)
- Java Development Kit (JDK)

Algorithm

1. Create a class named Student.
2. Declare student ID, name and age as data members.
3. Create three objects of the Student class.
4. Assign different values to each object.
5. Call the display() method for all three objects.

Program (Student.java)

```
class Student {
    int studentId;
    String name;
    int age;

    void display() {
        System.out.println("ID: " + studentId);
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println();
    }

    public static void main(String[] args) {
        Student s1 = new Student();
        s1.studentId = 101;
        s1.name = "Sameer";
        s1.age = 20;

        Student s2 = new Student();
        s2.studentId = 102;
        s2.name = "Rahul";
        s2.age = 21;

        Student s3 = new Student();
        s3.studentId = 103;
        s3.name = "Arjun";
        s3.age = 19;

        s1.display();
        s2.display();
        s3.display();
    }
}
```

Output (Attach Screenshot):

```
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>  & 'C:\Program Files\Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Anu
nuru\AppData\Roaming\Code\User\workspaceStorage\1e9d25f7c48886e17d4b2fa02b0347fc\redhat.java\jdt_ws\AOOP_c8922b60\bin' 'Student'
============================
     Student Details
============================
Student ID : 1
Name       : Anudeep
Age        : 20
----------------------------
Student ID : 2
Name       : Samrat
Age        : 22
----------------------------
Student ID : 3
Name       : Lachan
Age        : 21
----------------------------
```

Result
Thus, the Student class was created successfully, and the details of three student objects were displayed.

Experiment 2: Employee Class

Aim
To create an Employee class with employee ID, name and salary, initialize the values using a parameterized constructor and display the employee details.

Software Requirements
- MacBook / macOS
- Visual Studio Code (VS Code)
- Java Development Kit (JDK)

Algorithm

1. Create a class named Employee.

2. Declare employee ID, name and salary.

3. Define a parameterized constructor to receive the employee details.

4. Initialize the data members using the constructor arguments.

5. Create an Employee object and display its details.

Program (Employee.java)

```
class Employee {
   int empId;
   String name;
   double salary;

   Employee(int id, String n, double s) {
      empId = id;
      name = n;
      salary = s;
   }

   void display() {
      System.out.println("Employee ID: " + empId);
      System.out.println("Name: " + name);
      System.out.println("Salary: " + salary);
   }

   public static void main(String[] args) {
      Employee e1 = new Employee(101, "Sameer", 30000);
      e1.display();
   }
}
```

Output :

```
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>  & 'C:\Program Files\Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:
\Users\Anudeep_Potnuru\AppData\Roaming\Code\User\workspaceStorage\1e9d25f7c48886e17d4b2fa02b0347fc\redhat.java\jdt_ws\AOOP_c8922b60\bin' 'Empl
oyee'
Employee ID : 1
Name       : Digimon
Salary        : 100000
---------------------------
Employee ID : 2
Name       : Naruto
Salary        : 200000
---------------------------
Employee ID : 3
Name       : Sasuke
Salary        : 300000
---------------------------
```

Result

Thus, the Employee class was created successfully using a parameterized constructor, and the employee details were displayed.

Experiment 3: BankAccount Class

Aim

To create a BankAccount class with account number and balance and implement deposit and withdrawal methods.

Software Requirements

- MacBook / macOS
- Visual Studio Code (VS Code)
- Java Development Kit (JDK)

Algorithm

1. **Create a class named BankAccount.**
2. **Declare account number and balance.**
3. **Create a constructor to initialize the account details.**
4. **Create a deposit() method to add money to the balance.**
5. **Create a withdraw() method that checks whether sufficient balance is available.**
6. **Display the account number and final balance.**

Program (BankAccount.java)

```
class BankAccount {
    int accountNumber;
    double balance;

    BankAccount(int acc, double bal) {
        accountNumber = acc;
        balance = bal;
    }

    void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: " + amount);
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient Balance");
        }
    }

    void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {
```

```
        BankAccount b1 = new BankAccount(12345, 5000);
        b1.deposit(2000);
        b1.withdraw(1000);
        b1.display();
    }
}
```

Output

```
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>  & 'C:\Program Files\Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:
\Users\Anudeep_Potnuru\AppData\Roaming\Code\User\workspaceStorage\1e9d25f7c48886e17d4b2fa02b0347fc\redhat.java\jdt_ws\AOOP_c8922b60\bin' 'Bank
Account'
Deposited: 2000.0
Withdrawn: 1000.0
Account Number: 12345
Balance: 6000.0
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>
```

Result

Thus, the BankAccount class was created successfully, and deposit and withdrawal operations were performed.

Experiment 4: Rectangle Class

Aim
To create a Rectangle class with length and width and calculate its area and perimeter.

Software Requirements
- MacBook / macOS
- Visual Studio Code (VS Code)
- Java Development Kit (JDK)

Algorithm

1. **Create a class named Rectangle.**
2. **Declare length and width as data members.**
3. **Initialize the values using a constructor.**
4. **Calculate area using length × width.**
5. **Calculate perimeter using 2 × (length + width).**
6. **Display the calculated area and perimeter.**

Formula
Area = Length × Width
Perimeter = 2 × (Length + Width)

Program (Rectangle.java)

```
class Rectangle {
    double  length;
    double width;

    Rectangle(double l, double w) {
        length = l;
        width = w;
    }

    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    public static void main(String[] args) {
        Rectangle r1 = new Rectangle(10, 5);
```

```
        r1.calculate();
    }
}
```

Output:

```
PS C:\Users\Anudeep_Potnuru\Documents\AOOP> & 'C:\Program Files\Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:
\Users\Anudeep_Potnuru\AppData\Roaming\Code\User\workspaceStorage\1e9d25f7c48886e17d4b2fa02b0347fc\redhat.java\jdt_ws\AOOP_c8922b60\bin' 'Rect
angle'
Area: 50.0
Perimeter: 30.0
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>
```

Result

Thus, the Rectangle class was created successfully, and its area and perimeter were calculated.

Experiment 5: Book Class

Aim

To create a Book class with ISBN, title and price and initialize the values using a parameterized constructor.

Software Requirements

- MacBook / macOS
- Visual Studio Code (VS Code)
- Java Development Kit (JDK)

Algorithm

1. **Create a class named Book.**
2. **Declare ISBN, title and price.**
3. **Define a parameterized constructor to receive the book details.**
4. **Initialize the data members using the constructor arguments.**
5. **Create a Book object and display its details.**

Program (Book.java)

```
class Book {
    String ISBN;
    String title;
    double price;

    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {
        Book b1 = new Book("123456", "Java Programming", 500);
        b1.display();
    }
}
```

Output:

```
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>  & 'C:\Program Files\Java\jdk-20\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C
\Users\Anudeep_Potnuru\AppData\Roaming\Code\User\workspaceStorage\1e9d25f7c48886e17d4b2fa02b0347fc\redhat.java\jdt_ws\AOOP_c8922b60\bin' 'Boo
'
ISBN: 123456
Title: Java Programming
Price: 500.0
PS C:\Users\Anudeep_Potnuru\Documents\AOOP>
```

Result

Thus, the Book class was created successfully using a parameterized constructor, and the book details were displayed.